

MAC21 — The 48-Hour Plan

Now: Sunday 12:00 · **Exam:** Tuesday 09:30 · **Real study time available:** ~22 hours · **Target:** 60-70 marks by the easiest route

The whole strategy in one line

In each unit, answer **a) + b) + ONE of the two 7-markers**. That is $2 + 4 + 7 = 13$ marks per unit $\times 5$ units = **65 marks**.

You do not need both 7-markers. That halves the hardest part of the syllabus and is why this is achievable in two days. Anything you manage on the second 7-marker is bonus on top of 65.

The 5 questions you are answering

UNIT	ANSWER	A) — 2 MK	B) — 4 MK	YOUR ONE 7-MARKER	TOTAL
I	Q2	State Maclaurin's / Taylor's series	$xy(1-x-y)$ max at $(1/3, 1/3)$	Expand $xy+\cos(xy)$ about $(1, \pi/2)$ 4x	13
II	Q4	DE of L-C circuit / define orthogonal trajectories	Bungee Euler <i>or</i> Taylor's method <i>or</i> orthogonal trajectory	Modified Euler's method 5x (<i>sits at 4c in 4 of 5</i>)	13
III	Q5	Steps for solving Cauchy's LDE	Prove $1/(D+a) \cdot X$ 4x	Cauchy's equation 4x (<i>easier than variation of parameters</i>)	13
IV	Q8	Lagrange's inverse interpolation formula	Find the missing terms 3x	Newton's divided difference 5x <i>or</i> Simpson's 3/8	13
V	Q10	Rank / echelon / consistency definition	Gauss-Seidel, 2 iterations 3x	Gauss elimination word problem (<i>easier than system of ODEs</i>)	13
Floor if every 7-marker lands					65

What you are deliberately NOT studying. Every one of these lives in the question you are *not* answering. Touch them only if you finish everything else:

Legendre's equation (Q6) · **variation of parameters** (the other 7 in Q5 — learn only if time) · **Rayleigh's power method** (Q9) · **diagonalization / modal matrix** (Q9) · **Lagrange multipliers** (Q1) · **maxima-minima beyond $xy(1-x-y)$** (Q1) · **radius of curvature, Newton-Cotes derivation, backward difference tables.** That is roughly 40% of the syllabus, cut on purpose.

SUNDAY — 12:00 to 23:00 (~8.5 h study)

Today you bank **Units III, IV and V** — the three most predictable units, worth 39 marks. Do the hardest-to-fake unit while you are freshest.

TIME	BLOCK	WHAT EXACTLY
12:00–13:00	2-mark bank, pass 1	Section 6 of the prediction sheet. Write each definition out <i>by hand</i> , once. Do not just read. This is 10 marks and it is the cheapest hour of the whole weekend.
13:00–13:45	Lunch	Away from the desk.
13:45–16:00	UNIT III 13 mk	The single biggest win in the paper. • Memorise the $1/(D+a)$ proof cold — it is 4 marks, appeared 4x, and never changes. • Cauchy's equation: the substitution $x = e^z$, then it is a constant-coefficient problem. • Work M23 5c, M24 5d, MAM24 5d — three Cauchy problems. • Write the "steps for solving Cauchy's LDE" answer once.

16:00–16:20	Break	Walk. No screen.
16:20–18:30	UNIT V 13 mk	Smallest unit in the paper. - Gauss-Seidel — do the MP/P26 system ($x+8y+2z=-4$...), it appeared twice identically. - Then M23 9b and M24 10b. Three reps and it is automatic. - Gauss elimination: M23's electrical network and M24's dietician problem. - The five Unit V definitions (rank, echelon, consistency, modal & spectral, "is every matrix diagonalizable").
18:30–19:30	Dinner	Proper meal.
19:30–21:30	UNIT IV 13 mk	The most mechanical unit — good for a tired brain. - Missing terms — do the $45...65 / 3.0, -, 2.0, -, -2.4$ table, it appeared identically twice. - Newton's divided difference — the $(2,4)(4,56)(9,711)(10,980)$ cubic appeared twice verbatim. Do it. - Simpson's $3/8$ on $\int_0^1 dx/(1+x^2)$ as your backup 7-marker. - Lagrange's inverse interpolation formula — write it out.
21:30–22:30	Consolidate	Close the notes. From memory: the $1/(D+a)$ proof, one Cauchy problem, one Gauss-Seidel. Whatever you cannot do, mark it and fix it first thing tomorrow.
22:30–23:00	Wind down	No maths. Sleep by 23:00.

End of Sunday you should own: Units III, IV, V $a+b+one\ 7 = 39$ marks — plus most of the 2-mark bank.

MONDAY — 07:00 to 23:00 (~11.5 h study)

Units I and II in the morning, a timed mock at midday, repairs and bonus marks in the evening.

TIME	BLOCK	WHAT EXACTLY
07:00–07:30	Wake, breakfast	—
07:30–09:45	UNIT II 13 mk	Learn the four numerical methods <i>as one family</i> — they are variations of a single idea, and doing them together is far faster than one at a time. - Taylor's series method → Euler's → Modified Euler's → RK4. Same skeleton: tabulate, iterate, 4 decimals. Modified Euler is your 7-marker (slot 4c, 4 of 5 papers). Do three of them. - Newton's law of cooling — 10 minutes to learn, appeared 5x. Do the coffee problem. - Orthogonal trajectories, Cartesian and polar. The circuit-DE 2-marker.
09:45–10:00	Break	—
10:00–12:00	UNIT I 13 mk	Expand about $(1, \pi/2)$ — appeared 4x, the point never changes. Do all three variants: $xy+\cos(xy)$, $xy+\sin(xy)$, $xy^2+\cos(xy)$. $xy(1-x-y)$ maximum at $(1/3, 1/3)$ — appeared 3x. Learn the r-s-t test on it. - The Newton-Raphson system $x^2+y^2=x$, $x^2-y^2=y$ from $(0.8, 0.4)$ — appeared 4x. It is in Q1, but learn it: it is your escape hatch if Q2 looks bad. - The Unit I statements: Maclaurin one variable, Taylor two variables.
12:00–13:00	Lunch	—
13:00–15:00	TIMED MOCK critical	Sit the Model Paper. Answer Q2 · Q4 · Q5 · Q8 · Q10, $a + b + one\ 7$-marker each. Clock it: 2 hours. Closed book, no phone. This is the most important block of the two days — it converts "I know it" into "I can write it under time".
15:00–15:30	Break	Mark it honestly first, then rest.
15:30–17:30	Repair	Whatever you dropped in the mock, redo — the actual question, start to finish, not just "review the method". Anything you got right, do not touch again.
17:30–18:30	Dinner	—

18:30–20:30	BONUS 7-markers 65 → 80	Only now, with the floor secured, add the <i>second</i> 7-marker in each question: • Variation of parameters (Unit III, 5c/5d) — appeared 5×/5. Highest-value bonus. • System of ODEs by matrix method (Unit V, 10d) — 3×/3, always the final question. • Grouped-frequency estimation (Unit IV) — learn the cumulative-frequency setup. • RK4 (Unit II) — 6×/6, appears in every paper.
20:30–22:00	2-mark bank, pass 2	Write out all five units' definitions from memory. Then every formula you need: N-R, Modified Euler, RK4, Newton-Gregory, Lagrange, Simpson's 1/3 and 3/8, Gauss-Seidel, Cauchy substitution. One sheet, by hand.
22:00–23:00	<i>Wind down</i>	Sleep by 23:00. <i>Non-negotiable — see below.</i>

TUESDAY — 06:00 to 09:30

TIME	BLOCK	WHAT EXACTLY
06:00–06:40	<i>Wake, breakfast</i>	<i>Eat properly. Do not skip it.</i>
06:40–07:40	2-mark bank, final	All 5 units' definitions + your formula sheet, from memory. This is 10 marks and it is the last thing that should still be wobbling.
07:40–08:30	Setups only	For each of your five 7-markers, write only the <i>first three lines</i> — the setup, not the full solve. $1/(D+a)$ proof · Cauchy substitution · Modified Euler table · divided-difference table · Gauss-Seidel rearrangement. Ten minutes each.
08:30–08:50	Skim	Tier 1 table of the prediction sheet. Read, do not solve.
08:50	STOP	<i>Close everything. Travel. Learn nothing new after this point — it displaces what is already in place and costs more than it adds.</i>

In the exam hall — 09:30 to 12:30

MINUTES	DO THIS
0–5	Read all ten questions before writing anything. In each unit compare the two options and commit. Write your five choices on the top of the answer sheet so you do not drift.
5–20	All five 2-markers first. 10 marks in 15 minutes, while you are calm. Nothing else in the paper has that rate of return.
20–50	All five 4-markers. Another 20 marks. You are now at 30 with 100 minutes left.
50–170	The 7-markers — easiest first. 24 minutes each $\times 5 = 120$. Do your strongest unit first to build momentum. If one stalls past 25 minutes, leave it and move on; come back only if there is time.
170–180	Sweep. Check every question is numbered and lettered correctly. Attempt the second 7-marker anywhere you have a spare minute — partial marks are still marks.

Show every step. These are method-marked papers. A Gauss-Seidel table with an arithmetic slip in the last row still scores most of the 4. A correct final answer with no working does not. **Never leave a part blank** — write the formula and the setup even if you cannot finish; that is 2–3 marks on a 7-marker for ninety seconds of work.

Sleep 7 hours both nights. This is the highest-leverage instruction on the page. Studying until 2am costs you more in recall and arithmetic accuracy than the extra hours add — and this paper is almost entirely arithmetic accuracy. If you are behind on Monday night, cut the bonus 7-markers, not the sleep.

If you fall behind — the order to sacrifice things:

1. Bonus 7-markers (Monday evening) — drop entirely, you still have 65.

2. Unit I depth — keep only "expand about $(1, \pi/2)$ " and $xy(1-x-y)$.

3. Simpson's 3/8 backup in Unit IV — divided difference alone is enough.

Never sacrifice: the 2-mark bank, the $1/(D+a)$ proof, Gauss-Seidel, missing terms, Modified Euler. Those five are ~25 marks for maybe four hours of work.

Built from the MAC21 prediction sheet: 6 papers analysed, repeat counts stated out of the papers carrying each unit. Question-choice recommendations reflect observed slot frequency, not certainty — always read both options before committing. Timings assume you start at Sunday 12:00 and are asleep by 23:00 both nights.